

When should I suspect diabetic ketoacidosis in a person with type 1 diabetes?

Last revised in December 2024

- Suspect **diabetic ketoacidosis** (</topics/diabetes-type-1/background-information/complications/>) (DKA) in a person with known diabetes *or* significant hyperglycaemia (finger-prick blood glucose level greater than 11 mmol/L) and the following clinical features:
 - Increased thirst and urinary frequency.
 - Weight loss.
 - Inability to tolerate fluids.
 - Persistent vomiting and/or diarrhoea.
 - Abdominal pain.
 - Visual disturbance.
 - Lethargy and/or confusion.
 - Fruity smell of acetone on the breath.
 - Acidotic breathing — deep sighing (Kussmaul) respiration.
 - Dehydration, which can be classified as:
 - Mild — only just clinically detectable.
 - Moderate — dry skin and mucus membranes, and reduced skin turgor.
 - Severe — sunken eyes and prolonged capillary refill time.
 - Shock (resulting from severe dehydration). The person is severely ill with:
 - Tachycardia, poor peripheral perfusion, and (as a late sign) hypotension (indicating decreased cardiac output).
 - Lethargy, drowsiness, or decreased level of consciousness (indicating decreased cerebral perfusion).
 - Reduced urine output (indicating decreased renal perfusion).
- If DKA is suspected:
 - **Assess for precipitating factors of DKA, such as:**
 - Infection (for example pneumonia or a urinary tract infection).
 - Physiological stress (such as trauma or surgery).
 - Non-adherence to insulin treatment regimen or intentional insulin omission in order to lose weight ([diabulimia \(/topics/diabetes-type-1/background-information/complications/\)](/topics/diabetes-type-1/background-information/complications/)).
 - Other medical conditions (such as hypothyroidism or pancreatitis).
 - Drug treatment (such as corticosteroids, diuretics, and sympathomimetic drugs [for example salbutamol]).
 - **Test for ketones** (produced by the liver when there is a lack of glucose [starvation ketones] and as an alternative energy source when there is a relative insulin deficiency).
 - In an adult with suspected DKA, test for urine or blood ketones even if plasma glucose levels are near normal.
 - In a child or young person with suspected DKA, test for blood ketones. If this is not possible, arrange immediate admission to a hospital with acute paediatric

facilities.

- Ketones are high if above 2+ in the urine or above 3 mmol/L in the blood.
- **Consider the possibility of DKA in all people with type 1 diabetes who are unwell, bearing in mind that:**
 - Hyperglycaemia may not always be present — children and young people on insulin therapy may develop DKA with normal blood glucose levels.
 - Low blood ketone levels (less than 3 mmol/L) do not always exclude DKA.

Basis for recommendation

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